

Health Workforce Coverage and Gap Analysis for Selected Countries (African Region)

JKUAT Team

- Health workforce shortages remain a major barrier to achieving UHC.
- Many countries lack reliable, comparable estimates of the workforce required over time.
- Evidence-based projection tools help quantify gaps and test policy scenarios (BAU, Scale-Up, Shock).

Aim

Develop models to help quantify the workforce supply, demand and return on investment.

- Rapid population growth increases demand for health services,
- Persistent shortages in critical cadres constrain service delivery,
- Uneven cadre availability can limit progress toward UHC targets,
- Planning requires transparent models for workforce requirements, supply, coverage and gaps.

- Here, we consider a national Human Resources for Health (HRH) projection and gap analysis tool, tested using data from for five countries:
 - **Burkina Faso,**
 - **Democratic Republic of Congo (DRC),**
 - **Guinea-Bissau,**
 - **Mozambique and**
 - **South Africa,**but it can be extended in other countries.
- It estimates future health workforce supply, compares it against **alternative workforce requirement models**, and quantifies staffing gaps up to 2035 under different policy scenarios.

Defining Workforce Requirement

The workforce requirement is estimated using two complementary approaches:

- ① A **Benchmark** (normative density) model, and
- ② A **Need-linked** (burden-adjusted) model. *Adjusts workforce requirements based on health burden indicators.*

Let

t index year,

c index cadre, and

P_t denote the total population in year t .

1. Benchmark Requirement Model

For each cadre c , a target workforce density per 10,000 population, B_c , is specified. The required headcount for cadre c in year t is:

$$R_{c,t}^{bench} = \frac{B_c \times P_t}{10,000}$$

where:

- $R_{c,t}^{bench}$ = required stock under benchmark model
- B_c = benchmark density (per 10,000 population)
- P_t = population in year t

Assumption: workforce need grows proportionally with population.

2. Need-Linked Requirement Model (Step 1)

The need-linked model adjusts total required workforce density based on disease burden and health system performance indicators. The model is undertaken in steps.

Step 1: Standardization of need indicators

Let:

- $DALY_t$ = DALYs
- MMR_t = Maternal Mortality Ratio (MMR)
- NCD_t = NCD Mortality Rate
- UHC_t = UHC Service Coverage Index

Each indicator is standardized (different scales/units) using a z-score:

$$Z_{x,t} = \frac{X_t - \mu_X}{\sigma_X}$$

Composite need index:

$$NI_t = Z_{DALY,t} + Z_{MMR,t} + Z_{NCD,t} - Z_{UHC,t}$$

where higher UHC coverage reduces need.

After standardization, we combined the indicators using weights as follows;

$$NI_t = 0.25Z_{DALY,t} + 0.25Z_{MMR,t} + 0.25Z_{NCD,t} - 0.25Z_{UHC,t}$$

Need-Linked Requirement Model (Steps 2–4)

Step 2: Total required density

Let B_{base} denote the base total HRH density (per 10,000 population), and α the sensitivity parameter:

$$D_t = B_{base} \times (1 + \alpha NI_t)$$

α controls how strongly health need indicators influence the workforce requirement relative to the baseline density.

Step 3: Convert density to total required stock

$$R_t^{NL} = \frac{D_t \times P_t}{10,000}$$

Step 4: Allocation across cadres

Let s_c denote the share allocated to cadre c , such that:

$$\sum_c s_c = 1$$

Allocated: Doctors - 0.2, Nurses - 0.7, Pharmacists - 0.07, Dentists - 0.03

Cadre-specific need-linked requirement:

$$R_{c,t}^{NL} = s_c \times R_t^{NL}$$

Workforce supply evolves through:

$$S_{c,t} = S_{c,t-1}(1 + g_c)$$

where g_c is the estimated growth rate for cadre c .

- Growth rates reflect net changes from training/recruitment and attrition.
- Scenario assumptions (BAU, Scale-Up, Shock) modify the effective growth trajectories.

Coverage and Gap Metrics

For each cadre and year, supply $S_{c,t}$ is compared to requirement $R_{c,t}^{(k)}$, where $k \in \{\text{bench}, \text{need}\}$.

Coverage ratio:

$$\text{Coverage}_{c,t} = \frac{S_{c,t}}{R_{c,t}}$$

Workforce gap:

$$\text{Gap}_{c,t} = S_{c,t} - R_{c,t}$$

Coverage > 1 indicates surplus; coverage < 1 indicates shortage.

Key Model Quantities (Summary)

Required workforce for cadre c in year t under requirement type k is:

$$R_{c,t}^{(k)}$$

Benchmark:

$$R_{c,t}^{bench} = \frac{B_c \times P_t}{10,000}$$

Need-linked:

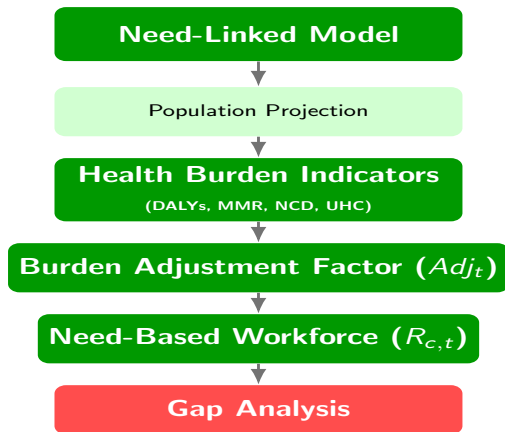
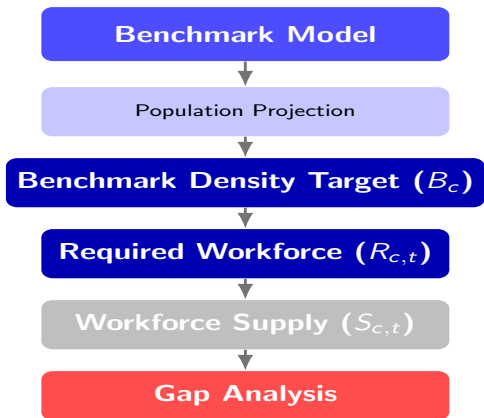
$$R_{c,t}^{NL} = s_c \times \frac{B_{base}(1 + \alpha NI_t) \times P_t}{10,000}$$

Supply:

$$S_{c,t} = S_{c,t-1}(1 + g_c)$$

- P_t = population in year t
- B_c = benchmark density per 10,000
- B_{base} = base total HRH density
- α = sensitivity to health need
- NI_t = composite need index
- s_c = cadre share
- g_c = cadre-specific growth rate

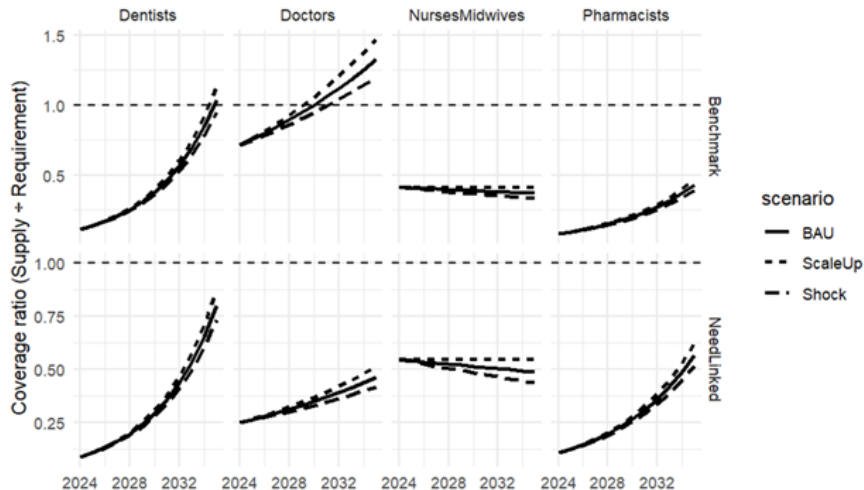
Health Workforce Requirement Models



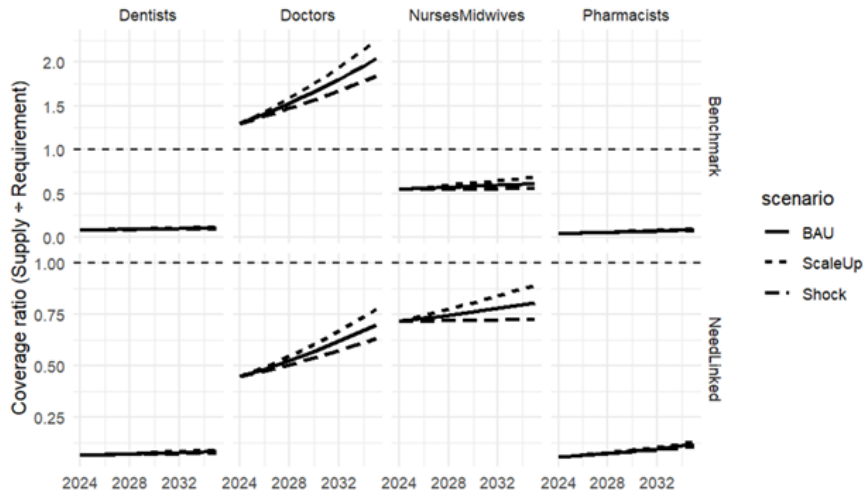
Scenario Comparison (Interpretation)

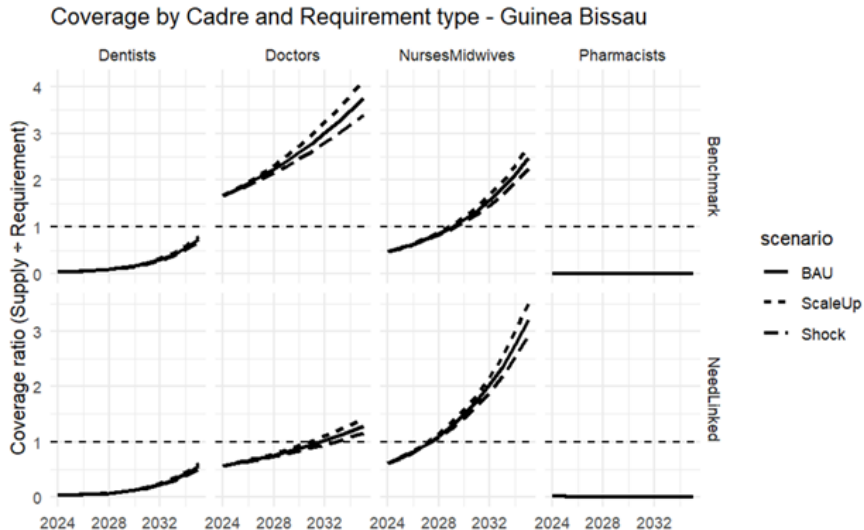
- **BAU**: continuation of current growth patterns.
- **Scale-Up**: accelerated training/recruitment and improved retention.
- **Shock**: reduced growth and/or increased attrition.
- Scenarios quantify resilience and time to close workforce gaps.

Coverage by Cadre and Requirement type - Burkina Faso

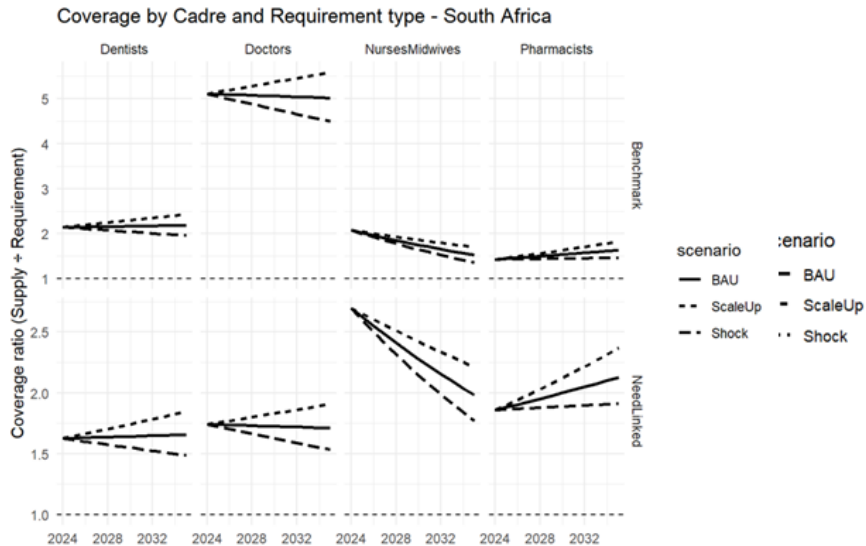


Coverage by Cadre and Requirement type - DRC





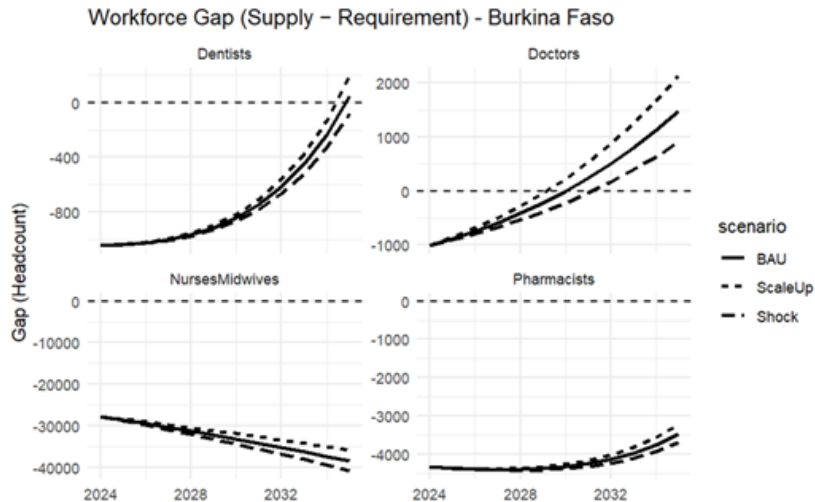
South Africa



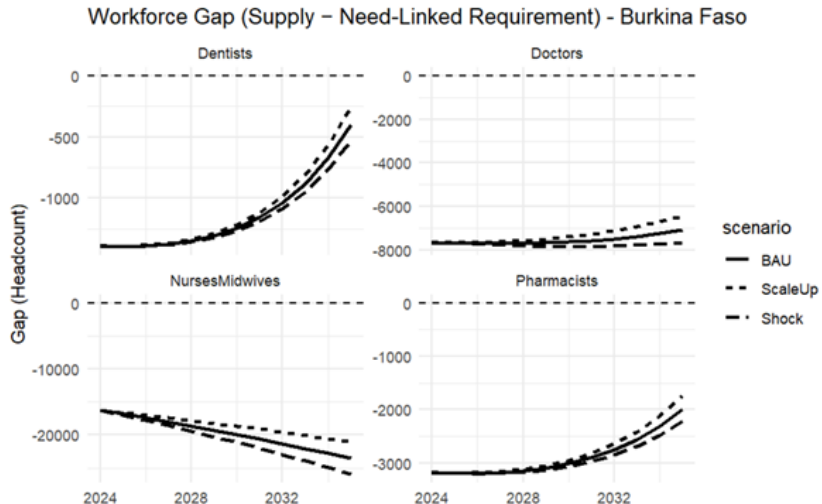
Coverage Analysis: Results Summary

- Across Guinea-Bissau, Mozambique, South Africa, the DRC, and Burkina Faso, workforce coverage generally improves over time, especially under Scale-Up scenarios.
- Coverage remains uneven across cadres, with persistent shortages among dentists and pharmacists in most countries.
- South Africa shows relative adequacy for several cadres.
- Need-linked requirements often produce larger shortfalls than benchmark targets, highlighting the magnitude of gaps when adjusted for population health need.

Burkina Faso: Gap (Benchmark Requirement)



Burkina Faso: Gap (Need-Linked Requirement)



Results - Burkina Faso

Gap analysis results are presented for Burkina Faso (but can be obtained similarly for all countries).

- Under the benchmark requirement, doctors move into surplus and dentists nearly close their gap, while nurses/midwives and pharmacists remain in large deficits.
- Under the need-linked requirement, all cadres remain below adequacy, especially nurses/midwives.
- Workforce shortages are more pronounced when adjusted for population health need.
- Scale-Up improves gaps, while Shock worsens them.

Cadre distribution gaps

- Coverage ratios can diverge substantially across cadres.
- Persistent under-coverage in selected cadres indicates cadre distribution gaps.
- Cadre distribution gaps can remain even when total workforce coverage improves.

Strengths

- Transparent requirement estimation using population and burden signals.
- Comparable framework across multiple countries.
- Supports scenario analysis for planning and resilience.

Limitations

- Dependent on quality and availability of workforce and burden data.
- Parameter choices (B_c , B_{base} , α , s_c) influence estimates.

Thank You